IAT New Test 16

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

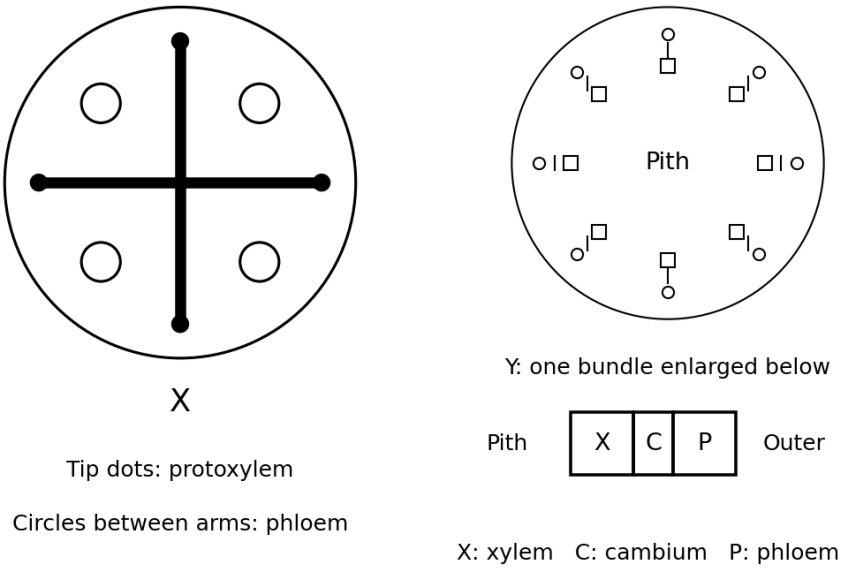
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 In Pinus, pollen reaches the ovule and a pollen tube delivers male gametes without a film of external water. The female gametophyte remains within the ovule. Which conclusion follows?
A) The female gametophyte must be a free-living prothallus
B) Pollen tubes prove that an ovary encloses the ovule
C) A non-flowering seed plant need not require external water for fertilisation
D) The ovule must leave the parent before fertilisation

Q2 In the figure, X has alternating xylem and phloem, with protoxylem at the outer tips of four xylem arms. Y has bundles in a ring; each has xylem toward the pith, phloem outward and a cambial strip between them. Which combined interpretation is correct?

Transverse vascular arrangements X and Y



A) X is a stem with endarch xylem; Y has closed bundles
B) X is a root with exarch xylem; Y can form secondary xylem toward the pith
C) X is a root with endarch xylem; Y forms secondary phloem toward the pith
D) X is a stem with radial bundles; Y lacks a lateral meristem

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Q3 A purified enzyme catalyses a reversible reaction in a closed tube. At the final plateau, labelled substrate still exchanges into product, but bulk concentrations remain constant. Adding more active enzyme does not change those final concentrations. Which interpretation fits?
A) The system is at dynamic equilibrium, whose composition is not changed by enzyme amount
B) Substrate has been completely exhausted
C) The original enzyme has become wholly inactive
D) The added enzyme favours only the reverse reaction

Q4 A tissue releases CO2 by alcoholic fermentation while taking up no oxygen. For this interval, how should respiratory quotient, defined as CO2 released divided by O2 consumed, be described?
A) Zero
B) Exactly one
C) Less than one
D) No finite numerical quotient can be assigned because its denominator is zero

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Q5 In a permeabilised muscle preparation at high Ca2+, removing ATP leaves many myosin heads attached to actin. ATP is then restored while free Ca2+ is strongly lowered. Which immediate sequence fits the cross-bridge mechanism?
A) ATP permits detachment; low Ca2+ limits renewed attachment by restoring site blocking
B) Low Ca2+ directly breaks all cross-bridges without ATP
C) ATP permanently exposes actin sites even after Ca2+ falls
D) Actin and myosin filaments must shorten before detachment

Q6 A monoecious plant has separate male and female flowers on one individual. A dioecious species has those flower types on different individuals. Under ordinary sexual reproduction, what is guaranteed by the second arrangement?
A) Only autogamy is prevented; geitonogamy remains possible
B) External pollen-transfer agents become unnecessary
C) Offspring become genetically identical to their mother
D) Both autogamy and geitonogamy are prevented

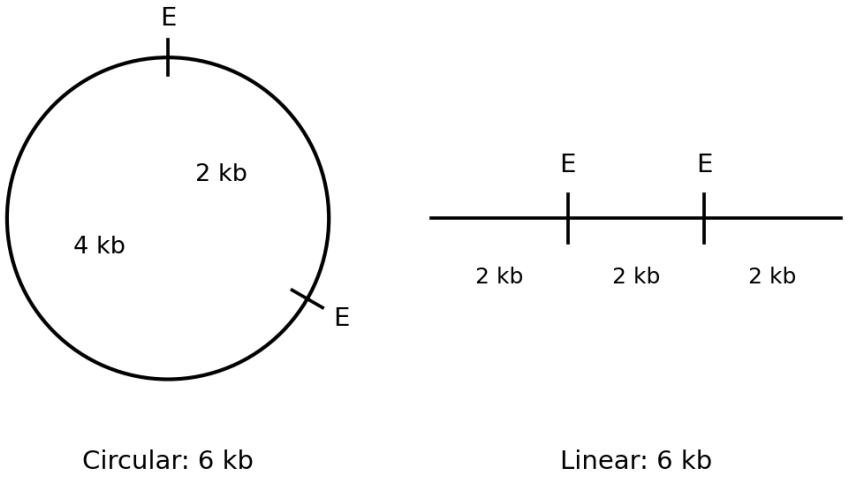
Q7 A coupling heterozygote AB/ab is test-crossed. Surviving progeny number 160 AB, 160 ab, 40 Ab and 40 aB. An independent assay shows each recombinant class has half the survival probability of either parental class. Treat the proportions as exact and recombination as equal in both sexes. Two heterozygotes of this same phase are later crossed with no viability differences. Which pair gives corrected recombination fraction and later aabb probability?
A) 1/5 and 4/25
B) 1/3 and 1/36
C) 1/3 and 1/9
D) 1/5 and 1/25

Q8 A drug blocks HIV reverse transcriptase but does not remove viral DNA already integrated into host chromosomes. It is added before infection in P and after integration in Q. Entry and cell viability are unaffected. Which prediction follows?
A) Both cultures immediately lose all viral genetic material
B) Only entry into P is prevented
C) Q cannot transcribe any integrated viral sequence
D) P is impaired in making new viral DNA; existing integrated templates in Q are not directly erased

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Q9 The figure gives a circular 6 kb DNA with two E sites separated by 2 kb and 4 kb arcs, and linear 6 kb DNA with E sites 2 kb and 4 kb from its left end. Both are completely digested; equal-size fragments migrate together. How many molecular fragments result from each?

Two E sites: circular versus linear DNA



Circular: 6 kb Linear: 6 kb

A) Circular 2; linear 3
B) Circular 2; linear 2
C) Circular 3; linear 3
D) Circular 3; linear 2

Q10 A closed population follows dN/dt=rN(1-N/K), with constant r,K. Its instantaneous growth is 8 individuals/month at N=20 and 12 individuals/month at N=60. What is K?
A) 80
B) 100
C) 120
D) 200

Q11 A carrier woman for an X-linked recessive allele has children with an unaffected man. Sexes are equally likely, births independent and penetrance complete. Given exactly one of their two children is affected, what is the probability that at least one child is a daughter?
A) 1/4
B) 1/2
C) 2/3
D) 3/4

Q12 A DNA population uniformly labelled with 15N in both strands undergoes exactly three synchronous semiconservative rounds in 14N medium, without DNA loss or further 15N. What fractions are hybrid duplexes before denaturation and heavy single strands after denaturation?
A) 1/8 and 1/8
B) 1/4 and 1/8
C) 1/4 and 1/4
D) 1/8 and 1/4

Q13 Cutting spinal root X abolishes a reflex response to skin stimulation, but direct stimulation of the intact motor root still contracts the muscle. In another preparation, cutting root Y leaves sensory impulses entering the cord but prevents contraction. Which assignment fits both?
A) X dorsal sensory; Y ventral motor
B) X ventral motor; Y dorsal sensory
C) Both exclusively sensory
D) Both exclusively motor

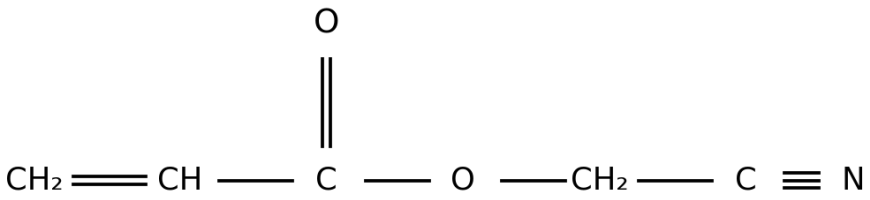
Q14 A ribosome takes exactly 30 s from initiation to complete-protein release. An mRNA initiates ribosomes at t=0,4,8,... s, without interference, degradation or premature termination. At t=20 s an inhibitor blocks new initiation before the event due then, without slowing existing ribosomes. At t=40 s how many proteins have been released and ribosomes remain translating?
A) 2 released; 3 translating
B) 3 released; 2 translating
C) 3 released; 3 translating
D) 5 released; none translating

Q15 A dicot-stem cambial strip produces derivatives on both sides. Mature derivatives on P contain vessel elements; Q includes sieve-tube elements and companion cells. Which interpretation is consistent?
A) P outward, secondary phloem
B) P inward, entirely living at maturity
C) P inward, secondary xylem; Q outward, including living conducting cells
D) Both products of apical meristem only

Chemistry

Q16 The displayed molecule is $CH_2{=}CH{-}C(=O){-}O{-}CH_2{-}C{\equiv}N$. Count all sigma and pi bonds, including C-H bonds, and classify only carbon-atom hybridisation. Which tuple gives (sigma bonds, pi bonds; sp, sp2, sp3 carbon counts)?

One connected multifunctional molecule



A) (7,4;1,3,1)
B) (12,4;1,3,1)
C) (12,3;2,2,1)
D) (12,4;1,2,2)

Q17 One mole of a mixture contains only $[Co(NH_3)_5Cl]Cl_2$ and $[Co(NH_3)_4Cl_2]Cl$. Excess AgNO3 immediately yields 1.60 mol AgCl; coordinated chloride is not displaced in this test. Assume complete dissociation of counterions. What are the mole fraction of the first compound and total moles of dissolved ions per initial mole of mixture?
A) 0.60 and 2.60
B) 0.40 and 2.40
C) 0.60 and 1.60
D) 0.80 and 3.20

Q18 Two Lewis proposals for CO are shown. P has three C-O shared pairs and one lone pair on each atom. Q has two shared pairs, one lone pair on C and two on O. Both use ten valence electrons. Which octet/formal-charge assessment is correct?

Lewis proposals P and Q



Each dot pair represents one lone pair

A) Both satisfy both octets and have identical formal charges
B) P violates oxygen's octet because oxygen has a positive formal charge
C) Q is preferred because carbon has an octet and oxygen has six electrons
D) P gives both octets with C=-1,O=+1; Q leaves carbon with only six electrons

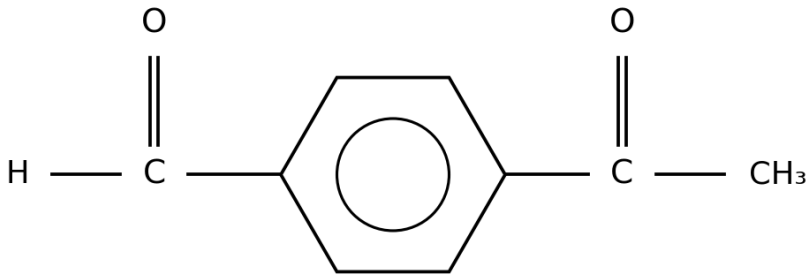
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Q19 For hydrogen, $1/\lambda = R(1/n_f^2 - 1/n_i^2)$. Let λ_B be the Balmer-series limiting wavelength and λ_L the Lyman-alpha wavelength (2 to 1). What is λ_B/λ_L?
A) 1/3
B) 4/3
C) 3
D) 4

Q20 In a Dumas nitrogen analysis, 2.80 g of a pure compound yields 224 mL of dry N2 at conditions where molar gas volume is 22.4 L/mol. All its nitrogen is recovered as N2. Its molar mass is 280 g/mol. How many nitrogen atoms are present per molecule?
A) 1
B) 4
C) 10
D) 2

Q21 The displayed compound is para-acetylbenzaldehyde, $OHC{-}C_6H_4{-}COCH_3$. Excess NaBH4 reduces both carbonyls, leaving the ring unchanged; both faces of the ketone react without optical bias. On separate portions of the worked-up product, what are the Tollens test, iodoform test and maximum configurational-stereoisomer count?

Reactant connectivity



A) Positive; positive; 2
B) Negative; negative; 1
C) Negative; positive; 2
D) Negative; positive; 4

Q22 Three amines undergo the Hinsberg test. P forms a sulfonamide soluble in aqueous alkali; Q forms a sulfonamide insoluble in alkali; R forms no sulfonamide but dissolves in dilute acid. Which class assignment is consistent under the standard test conditions?
A) P secondary; Q primary; R tertiary
B) P primary; Q secondary; R tertiary
C) P tertiary; Q secondary; R primary
D) P primary; Q tertiary; R secondary

Q23 Complete hydrolysis of a nucleic-acid sample yields phosphate, a pentose identified as ribose, and bases including uracil. Which interpretation uses both organic components?
A) It is consistent with RNA rather than ordinary DNA
B) It must be DNA because phosphate was found
C) It must be a protein because a nitrogenous base was found
D) It must be a reducing disaccharide

Q24 A first-order reaction runs for 10 min at T1 and then 5 min at T2. The rate constant at T2 is twice that at T1. The final concentration is one quarter of the initial concentration. What is its half-life at T1?
A) 5 min
B) 20 min
C) 40 min
D) 10 min

Q25 A solution initially contains 0.010 M I- and 0.10 M Cl-. Ag+ is added slowly with negligible volume change. Take Ksp(AgI)=10^-16 and Ksp(AgCl)=10^-10. Ignore complexation and use equilibrium free-ion concentrations. Which solid appears first, and what is [I-] when AgCl just begins to precipitate?
A) AgI; 10^-7 M
B) AgI; 10^-5 M
C) AgCl; 10^-7 M
D) AgCl; 10^-2 M

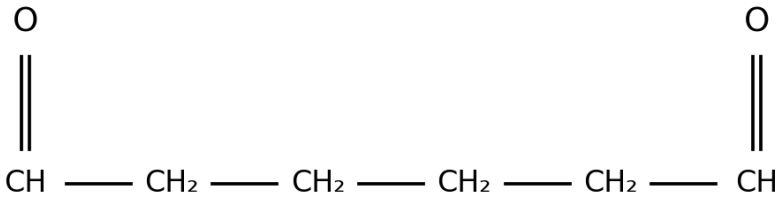
Q26 In acidic solution, iodate and iodide react completely to form iodine: $IO_3^- + I^- + H^+ \longrightarrow I_2 + H_2O$. For 0.020 mol iodate, which pair gives required iodide and produced iodine?
A) 0.020 mol I-; 0.020 mol I2
B) 0.100 mol I-; 0.060 mol I2
C) 0.120 mol I-; 0.060 mol I2
D) 0.100 mol I-; 0.120 mol I2

Q27 Estimate the enthalpy of $C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$ from average bond enthalpies in kJ/mol: C=C 612, H-H 436, C-C 348 and C-H 413. What is the estimate?
A) +126 kJ/mol
B) +222 kJ/mol
C) -126 kJ/mol
D) -539 kJ/mol

Q28 An ideal binary liquid has pure vapour pressures p*A=60 kPa and p*B=20 kPa. At one composition the measured partial pressures are pA=30 kPa and pB=10 kPa. What are the liquid and vapour mole fractions of A, respectively?
A) 1/2 and 1/2
B) 3/4 and 1/2
C) 3/4 and 3/4
D) 1/2 and 3/4

Q29 Reductive ozonolysis of a hydrocarbon containing one C=C gives the single molecular product drawn, $OHC-(CH_2)_4-CHO$, without breaking other carbon-carbon bonds. Which parent fits the full connectivity?

The single reductive-ozonolysis product



A) Cyclohexene
B) Hex-1-ene
C) Hexa-1,5-diene
D) Cyclopentene

Q30 Compare neopentyl bromide, $(CH_3)_3C-CH_2Br$, with 1-bromopropane. For a concerted E2 pathway and ordinary backside SN2 attack, which statement about neopentyl bromide is correct?
A) It has many beta hydrogens because its methyl groups contain hydrogen
B) It has no beta hydrogen, while branching next to its primary reaction centre hinders SN2
C) It must give 2-methylpropene by direct E2 without rearrangement
D) Its primary carbon guarantees faster SN2 than all unbranched primary halides

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Mathematics

Q31 All real normals from (6,0) to the parabola $y^2 = 4x$ touch it at points P_i. What is the sum of the x-coordinates of the distinct P_i?
A) 8
B) 0
C) 4
D) 14

Q32 A hyperbola centred at the origin has transverse axis along x and asymptotes y=2x and y=-2x. It passes through (3,4). What is the distance between its foci?
A) 5
B) $2\sqrt{5}$
C) $4\sqrt{5}$
D) 10

Q33 For real a let $f_a(x) = |x^3 - 3x + a|$. For which a does f have exactly one real point of non-differentiability?
A) $(-\infty, -2) \cup (2, \infty)$
B) $[-2, 2]$
C) $(-\infty, -2] \cup [2, \infty)$
D) $(-2, 2)$

Q34 Evaluate $I = \displaystyle\int_0^1 \frac{\ln(1+x)}{\ln(1+x) + \ln(2-x)}\, dx$.
A) 1
B) 1/2
C) ln2
D) 0

Q35 On the real numbers define xRy exactly when cos x=cos y. What is the full equivalence class containing $\pi/3$?
A) $\{2k\pi + \pi/3 : k \in \mathbb{Z}\}$
B) $\{k\pi + \pi/3 : k \in \mathbb{Z}\}$
C) $\{2k\pi \pm \pi/3 : k \in \mathbb{Z}\}$
D) $\{2k\pi \pm 2\pi/3 : k \in \mathbb{Z}\}$

Q36 Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. A real matrix X satisfies $AX - XA = \begin{pmatrix} 0 & 2 \\ -3 & 0 \end{pmatrix}$, trace X=5 and det X=0. What are all possible values of its upper-left entry?
A) -1 and 6
B) 2 and 3
C) 0 and 5
D) 1 and 4

Q37 Three independent identical trials have success probability p with 0<p<1. Exactly one success and exactly two successes are equally likely. Given at least one success, what is the probability of an even number of successes?
A) 3/7
B) 3/8
C) 1/2
D) 4/7

Q38 The coefficient of x^3 in $(1+ax)^5$ is 80, with a real. What is the sum of all coefficients of $(1-ax)^4$?
A) 81
B) 1
C) 16
D) -1

Q39 An increasing arithmetic progression has sum 40 in its first five terms and sum 115 in its sixth through tenth terms. What is the least positive n for which the nth term exceeds 40?
A) 12
B) 13
C) 14
D) 15

Q40 Let sgn denote the sign function. What is $\lim_{x\to 0^+} x\,\mathrm{sgn}(\sin(1/x))$?
A) 1
B) -1
C) Does not exist
D) 0

Q41 Lines $L_1 = (t, t, 0)$ and $L_2 = (1 + s, -s, 2)$ extend infinitely. What is the midpoint of the shortest segment joining them?
A) (1/2,1/2,1)
B) (1/2,-1/2,1)
C) (1,0,1)
D) (0,0,1)

Q42 Six distinct people A,B,C,D,E,F sit around a circular table. The shorter route from A to B around the table passes exactly one occupied seat, so their seat separation is two steps. C must not sit adjacent to A. Rotations are identical, reflections distinct. How many arrangements satisfy both conditions?
A) 12
B) 36
C) 48
D) 24

Q43 For $|z| = 1$, use principal arguments in $(-\pi, \pi]$. For which $\theta = \mathrm{Arg}\, z$ does $\mathrm{Arg}(z^2) = 2\,\mathrm{Arg}\, z - 2\pi$ hold?
A) $(0, \pi/2]$
B) $(-\pi, -\pi/2)$
C) $(\pi/2, \pi]$
D) $[-\pi/2, \pi/2]$

Q44 Solve $y' = y(1 - y)$, $y(0) = 2$. Which pair gives $y(\ln 2)$ and the maximal real interval of this solution containing zero?
A) $4/3; \mathbb{R}$
B) $2/3; (-\ln 2, \infty)$
C) $4/3; (-\infty, \ln 2)$
D) $4/3; (-\ln 2, \infty)$

Q45 Using principal arcsin in $[-\pi/2, \pi/2]$ and arccos in $[0, \pi]$, find all real x satisfying $\arcsin \frac{2x}{1+x^2} + \arccos \frac{1-x^2}{1+x^2} = \pi$.
A) $[1, \infty)$
B) $(1, \infty)$
C) $(-\infty, -1] \cup [1, \infty)$
D) $[0, 1]$

Physics

Q46 A projectile launched and landing at the same level has maximum height 5.0 m. On its upward path at height 3.0 m, its velocity makes 45 degrees with the horizontal. Neglect air resistance; g=10 m/s2. What is its horizontal range?
A) $2\sqrt{10}$ m
B) 20 m
C) 10 m
D) $4\sqrt{10}$ m

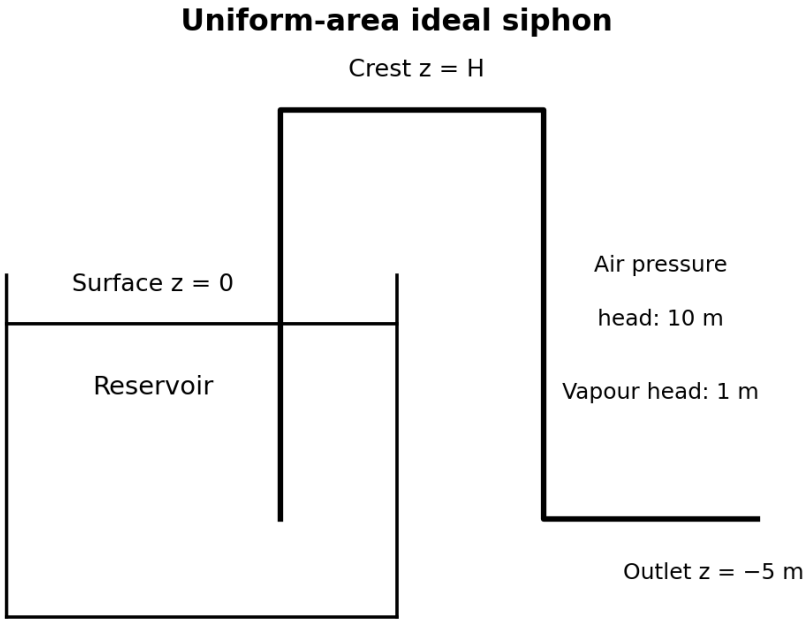
Q47 A 3.0 kg block rests on a horizontal floor. A separate vertical upward force of 6.0 N acts on it, without lifting it. The coefficient of static friction is 0.50. With g=10 m/s2, what horizontal force just reaches impending slip?
A) 15 N
B) 18 N
C) 12 N
D) 30 N

Q48 A 2.0 kg cart is driven by an engine delivering constant mechanical power 36 W. A constant 3.0 N resistance opposes forward motion. What are its terminal speed and forward acceleration when it moves at half that terminal speed?
A) 6 m/s; 1.5 m/s2
B) 12 m/s; 1.5 m/s2
C) 12 m/s; 3 m/s2
D) 6 m/s; 3 m/s2

Q49 Two identical coaxial discs, each moment of inertia I, initially rotate at +omega and -2omega. They couple and stick with no external axial torque. What are their final angular velocity and fraction of initial rotational kinetic energy lost?
A) -omega; 4/5
B) +omega/2; 9/10
C) -omega/2; 9/10
D) -omega/2; 1/10

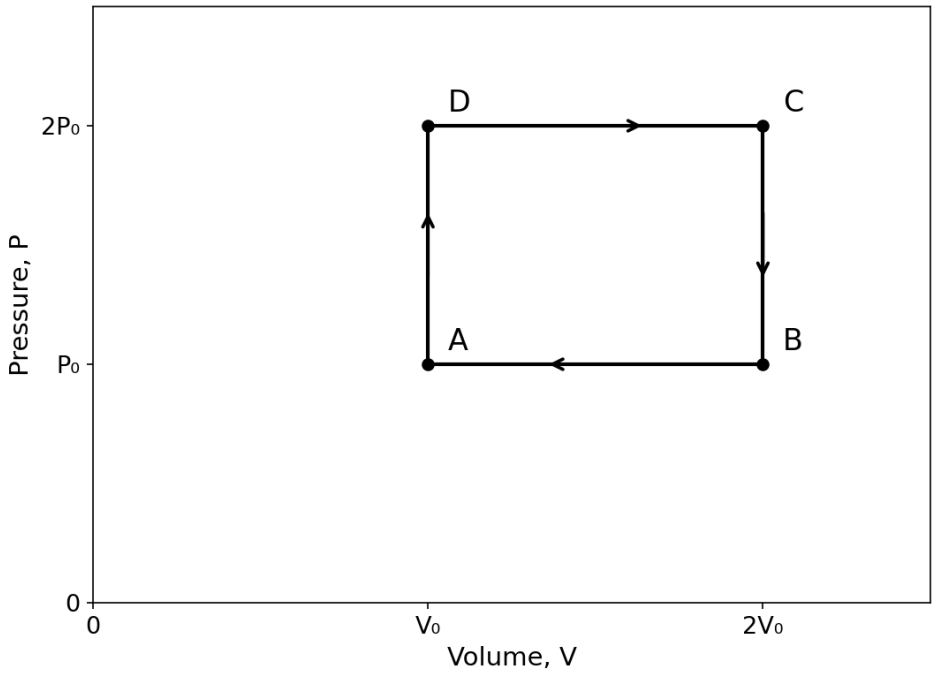
Q50 A satellite has a circular orbit just above a spherical planet's surface, with orbital period T. Neglect atmosphere and rotation. In terms of G and T, what is the planet's mean density?
A) $\pi/(GT^2)$
B) $3\pi/(GT^2)$
C) $3\pi/(2GT^2)$
D) $4\pi/(GT^2)$

Q51 An ideal steady siphon of uniform tube area draws from a large open reservoir. Its outlet is 5.0 m below the free surface and its crest H above it, as shown. Atmospheric pressure is equivalent to a 10.0 m water head; vapour pressure to 1.0 m. Ignore losses and free-surface speed. What limiting H brings crest pressure just to vapour pressure?



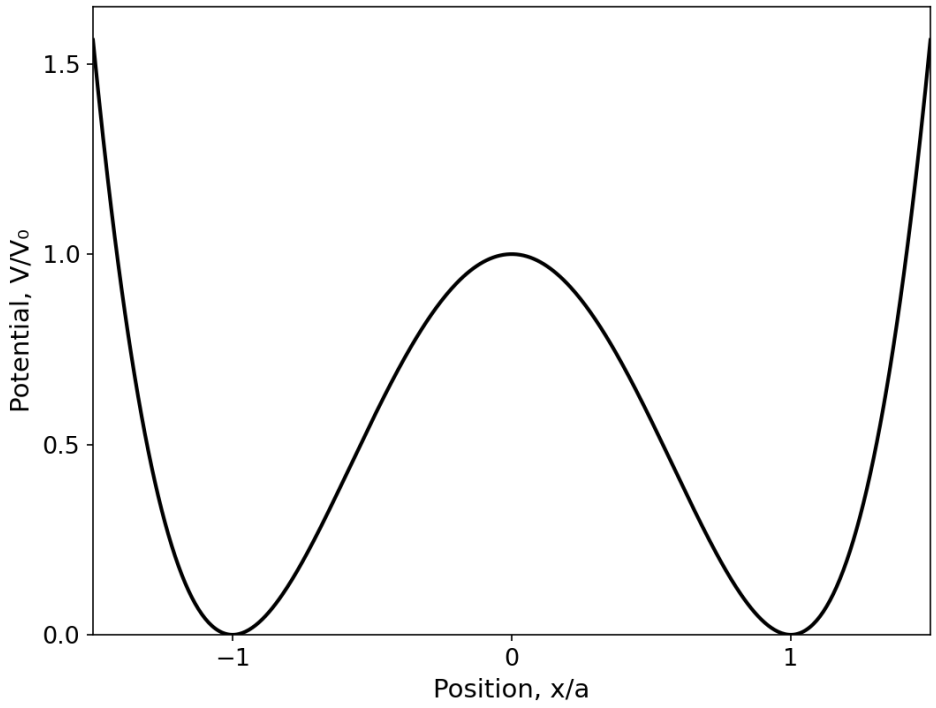
A) 5 m
B) 9 m
C) 4 m
D) 10 m

Q52 One mole of monatomic ideal gas undergoes the rectangular cycle A-D-C-B-A drawn in the P-V plane: A=(V0,P0), D=(V0,2P0), C=(2V0,2P0), B=(2V0,P0). All legs are quasistatic; Cv=3R/2 and P0V0=100 J. What is net work divided by total positive heat input?



A) 1/5
B) 4/13
C) 2/9
D) 2/13

Q53 The plotted electrostatic potential is $V(x) = V_0[(x/a)^2 - 1]^2$, with V0,a positive. A negative test charge is free to move along x only. At which finite equilibrium positions is it stable against small x displacements?



A) x=0 only
B) x=+a and -a only
C) x=0,+a,-a
D) None

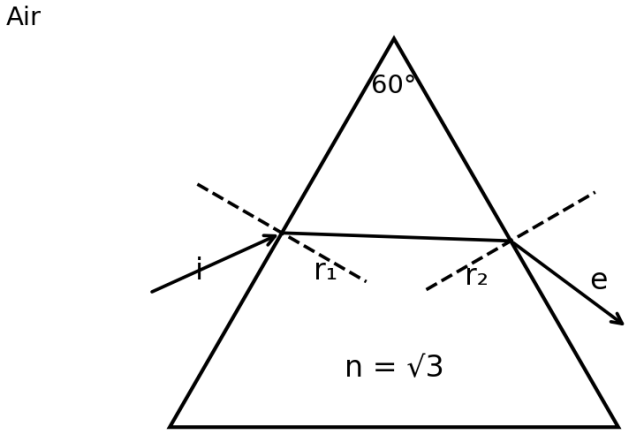
Q54 A capacitor C is at voltage V, and a capacitor 2C is at voltage V/2. Both sources are removed. The positive plate of each is joined to the negative plate of the other through resistive wires. After transients die, what is the magnitude of their common voltage?
A) V/2
B) 0
C) V/3
D) 2V/3

Q55 Two concentric coplanar circular loops have radii R and 2R. The first carries current I with magnetic moment positive along z. The second carries an opposing current J. If the net field at their common centre is zero, what are J and total magnetic moment?
A) 2I; $-7I\pi R^2\hat{z}$
B) 2I; zero
C) I/2; $-I\pi R^2$
D) 4I; $-15I\pi R^2\hat{z}$

Q56 A straight rod of length 3a rotates counterclockwise about a pivot a from end A and 2a from end B. Its angular speed is omega about +z and a uniform magnetic field B points along -z. Neglect contact effects. What is the motional-emf magnitude between ends, and which end is at higher potential?
A) $9B\omega a^2/2$; A
B) $3B\omega a^2/2$; B
C) $5B\omega a^2/2$; A
D) $3B\omega a^2/2$; A

Q57 A prism in air has apex angle 60 degrees and refractive index sqrt3. A ray enters one face; its internal angles to the two normals obey r1+r2=60 degrees. What are the limiting sin(i) below which the second face gives total internal reflection, and the minimum deviation for an emergent ray?

Prism angles are measured from normals



A) $(\sqrt{6}-1)/2;\ 60°$
B) $1/2;\ 60°$
C) $(\sqrt{6}-1)/2;\ 30°$
D) $(\sqrt{6}+1)/2;\ 60°$

Q58 Unpolarised light of intensity I0 meets two ideal crossed polarisers. A third ideal polariser is inserted between them at adjustable angle theta to the first. What is the maximum final intensity and one theta attaining it?
A) 0; every theta
B) I0/8; 45 degrees
C) I0/4; 45 degrees
D) I0/2; 90 degrees

Q59 Hydrogen atoms initially at n=2 meet single 2.00 eV photons. Relevant energies are E2=-3.40 eV, E3=-1.51 eV, E4=-0.85 eV; higher bound levels lie above E4. Neglect line widths, collisions and multiphoton processes. What happens in resonant electronic absorption?
A) They all reach n=3 because 2.00 exceeds 1.89 eV
B) They ionise because photon energy is positive
C) No listed or higher bound transition matches; the photons do not cause resonant excitation
D) They reach n=4 by rounding the energy gap

Q60 Two deuterons fuse at negligible initial kinetic energy to form helium-3 and a neutron. Binding energies per nucleon are 1.10 MeV for deuterium and 2.60 MeV for helium-3. Neglect recoil mass differences except that final masses are in ratio 3:1. What is the neutron kinetic energy in the centre-of-mass frame?
A) 3.40 MeV
B) 0.85 MeV
C) 1.50 MeV
D) 2.55 MeV

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / C Pollen-mediated fertilisation with a retained female gametophyte

Answer: (C) External water is not required for fertilisation in this non-flowering seed plant.

Why: Pinus is a gymnosperm, so absence of flowers does not imply the water requirement familiar from bryophytes and pteridophytes. Pollen transports the male gametophyte and its tube provides a delivery route. Retention of the female gametophyte also differs from a free-living fern prothallus; neither observation establishes an enclosing ovary.

Why not the others: (A) contradicts retention within the ovule; (B) confuses a pollen tube with an angiosperm ovary; (D) reverses retention of the ovule on the sporophyte.

Key point: Separate pollen-mediated fertilisation from flowers and fruits.

Q2 / B Root exarchy versus open stem bundles and derivatives

Answer: (B) X is an exarch root; Y can form secondary xylem inward.

Step 1: X has xylem and phloem on different radii, identifying radial root organisation in this comparison.

Step 2: Outer protoxylem and later-formed xylem inward constitute exarch development, not the endarch pattern of a stem.

Step 3: Y has open conjoint bundles. Cambium forms secondary xylem inward and secondary phloem outward.

Why: Arrangement, maturation direction and meristem position are independent clues that must agree. Rotating the drawing does not change which side faces the organ centre. Secondary-tissue potential follows from cambium, not merely from a ring of bundles.

Why not the others: (A) reverses X's maturation and ignores cambium; (C) reverses maturation and cambial output; (D) misidentifies radial root anatomy and overlooks the lateral meristem.

Key point: Combine bundle arrangement, protoxylem position and cambial derivatives.

Q3 / A Isotopic exchange at catalytic equilibrium

Answer: (A) Dynamic equilibrium with enzyme-independent equilibrium composition.

Why: Isotope exchange demonstrates continuing molecular conversion, so the plateau is not proof of stopped catalysis. Forward and reverse rates can be equal while concentrations stay constant. More enzyme accelerates both directions without changing the equilibrium constant at the same temperature, and therefore does not shift the final composition.

Why not the others: (B) cannot explain exchange from remaining substrate; (C) contradicts continuing catalysis; (D) invents a selective reverse action for the same enzyme preparation.

Key point: Enzymes change approach rates, not equilibrium position.

Q4 / D Undefined RQ during fermentation

Answer: (D) No finite numerical quotient can be assigned.

Why: The numerator is positive but oxygen uptake is zero. Division by zero cannot yield a finite quotient; calling a process anaerobic does not supply a denominator. RQ describes gas exchange, not ATP yield or the fraction of living cells.

Why not the others: (A) would require zero CO2 with positive oxygen uptake; (B) assumes equal nonzero gas volumes; (C) presumes a positive oxygen uptake exceeding CO2 release.

Key point: Check whether a physiological ratio is defined before interpreting it.

Q5 / A ATP detachment combined with calcium-dependent blocking

Answer: (A) ATP permits detachment; low Ca2+ limits renewed attachment.

Why: ATP binding promotes release of an attached myosin head. Separately, falling calcium changes the troponin-tropomyosin regulation so actin sites become less accessible. The intervention addresses two different stages: release of existing attachments and prevention of sustained cycling.

Why not the others: (B) assigns ATP-dependent detachment directly to calcium removal; (C) confuses the energy substrate with the calcium-sensitive regulatory system; (D) contradicts sliding-filament contraction, in which filament lengths remain essentially unchanged.

Key point: Distinguish cross-bridge detachment from exposure of binding sites.

Q6 / D Dioecy versus the same-plant pollen route

Answer: (D) Both autogamy and geitonogamy are prevented.

Why: Separate sexual flowers already prevent fertilisation within the same flower. A monoecious individual can nevertheless receive pollen from another flower on itself, which is geitonogamy. Dioecy eliminates that route too because an individual lacks the other sexual flower type, requiring transfer between plants.

Why not the others: (A) overlooks the absence of both sexual flower types on one dioecious plant; (B) reverses the need for transfer; (C) confuses sexual outcrossing with clonal reproduction.

Key point: Geitonogamy is between flowers but genetically self-pollination.

Q7 / C Viability-corrected linkage and subsequent gamete union

Answer: (C) 1/3 and 1/9.

Given: Recombinant survival is half parental survival; later offspring survive equally.

Formula/Principle: Divide observed counts by relative survival before inferring gametes.

Step 1: Corrected relative outputs are 160,160,80,80, totalling 480.

Step 2:

$$r = (80 + 80)/480 = 1/3, \qquad P(ab) = 160/480 = 1/3.$$

Step 3:

$$P(aabb) = P(ab)^2 = (1/3)^2 = 1/9.$$

Why: A test-cross reports gametes directly only if survival does not bias the classes. Restoring the depleted classes changes both r and the parental-gamete frequency. The later cross explicitly removes that bias, so corrected gametes determine its zygote probability.

Why not the others: (A) uses uncorrected r and ab frequency; (B) wrongly assigns r/2 to parental ab; (D) uses uncorrected r and then mistakes it for ab frequency.

Key point: Correct sampling bias before inferring inheritance.

Q8 / D Reverse-transcriptase inhibition before versus after integration

Answer: (D) DNA formation is impaired in P; existing integrated DNA in Q is not erased.

Why: Reverse transcriptase copies incoming viral RNA into DNA during infection. Blocking it prevents that step, rather than undoing an earlier integration event. An integrated DNA template is therefore a different molecular state from incoming RNA, and timing changes what the inhibitor can directly prevent.

Why not the others: (A) invents destruction of existing DNA; (B) contradicts unaffected entry; (C) equates reverse transcription with transcription from an existing DNA template.

Key point: Locate an inhibitor's step and distinguish prevention from removal.

Q9 / A Topology-dependent fragment count versus band multiplicity

Answer: (A) Two circular-DNA fragments and three linear-DNA fragments.

Why: Two cuts divide the circle into its two intervening arcs, producing 2 kb and 4 kb pieces. Two internal cuts in a linear molecule produce three 2 kb pieces because the original free ends also bound segments. Equal fragments can share one gel position without becoming one molecular fragment.

Why not the others: (B) counts sites instead of linear segments; (C) applies cuts-plus-one to a circle; (D) makes both errors.

Key point: Fragment number differs from visible-band number.

Q10 / B Inferring carrying capacity from two rates

Answer: (B) 100 individuals.

Given: The same logistic parameters apply at both sizes.

Formula/Principle: Per-capita growth equals r-(r/K)N.

Step 1:

$$r - 20r/K = 8/20 = 0.4, \qquad r - 60r/K = 12/60 = 0.2.$$

Step 2:

$$40r/K = 0.2 \Rightarrow r/K = 0.005.$$

Step 3:

$$r = 0.4 + 20(0.005) = 0.5 \text{ month}^{-1}, \quad K = 0.5/0.005 = 100.$$

Why: The larger population grows faster in total but slower per individual. Dividing by N reveals the density dependence. K is where per-capita growth reaches zero, not the size with the largest observed increase.

Why not the others: (A) gives per-capita predictions inconsistent with both points; (C) likewise cannot fit the two rates with a common r; (D) doubles the inferred intercept by overlooking the N=20 offset.

Key point: Linearise logistic data using per-capita growth.

Q11 / C Family-level conditioning in an X-linked cross

Answer: (C) 2/3.

Given: Each birth has affected-son probability 1/4, unaffected-son probability 1/4 and daughter probability 1/2.

Formula/Principle: Condition on the full two-child event.

Step 1:

$$P(\text{exactly one affected}) = 2(1/4)(3/4) = 3/8.$$

Step 2:

$$P(\text{affected and daughter}) = 2(1/4)(1/2) = 1/4.$$

Step 3:

$$P(\text{daughter present} \mid \text{one affected}) = (1/4)/(3/8) = 2/3.$$

Why: The affected child must be a son, so the daughter condition concerns the other child, already known to be unaffected. Daughters constitute two of the three equally likely unaffected genotype/sex classes. Birth-order multiplicity cancels between numerator and denominator.

Why not the others: (A) gives the unconditional joint event; (B) ignores conditioning; (D) gives a single child's unaffected probability.

Key point: Apply the family observation before using a sex ratio.

Q12 / B Label conservation with duplex versus strand denominators

Answer: (B) 1/4 and 1/8.

Given: Start from one heavy-heavy duplex; newly made strands are light.

Formula/Principle: Original heavy strands persist while total duplex count doubles.

Step 1:

$$N_{\text{duplex}} = 2^3 = 8, \quad N_{\text{heavy strands}} = 2.$$

Step 2: Each original heavy strand is in a hybrid, so two of eight duplexes are hybrid.

Step 3:

$$f_{\text{hybrid}} = 2/8 = 1/4, \quad f_{\text{heavy single}} = 2/(2 \times 8) = 1/8.$$

Why: A hybrid contains only one heavy strand. After denaturation the denominator counts both strands of every duplex. Semiconservative copying conserves the original labelled chains rather than producing new heavy strands each generation.

Why not the others: (A) uses the single-strand fraction twice; (C) uses the duplex fraction twice; (D) interchanges the counting units.

Key point: Track labelled chains separately from their containing molecules.

Q13 / A Reciprocal spinal-root cuts and bypass stimulation

Answer: (A) X is dorsal sensory; Y is ventral motor.

Why: A reflex needs afferent input and efferent output. Direct motor-root stimulation bypasses X's cut and verifies the downstream output and muscle, localising the first failure to input. In the second preparation, sensory arrival verifies input while failed contraction identifies the interrupted output.

Why not the others: (B) reverses afferent and efferent roles; (C) cannot explain intact sensory arrival after Y's cut; (D) cannot explain the motor bypass of X.

Key point: A bypass control separates input failure from output failure.

Q14 / B Finite polysome cohort after initiation arrest

Answer: (B) Three released; two translating.

Given: Initiation occurs at 0,4,8,12,16 s only, with a 30 s transit.

Formula/Principle: Follow the already initiated cohort independently of initiation blockade.

Step 1: Five events occur strictly before 20 s.

Step 2:

$$t_{\text{release}} = 30, 34, 38, 42, 46 \text{ s}.$$

Step 3: At 40 s the first three times have passed; the last two have not.

Why: Blocking initiation does not stop elongation or instantly release unfinished chains. The timing stipulation explicitly excludes the event at 20 s, leaving five chains to follow. A polysome can therefore continue releasing proteins after no new ribosomes enter.

Why not the others: (A) misses the 38 s release; (C) wrongly permits initiation at 20 s; (D) confuses blocking entry with finishing every chain.

Key point: Separate initiation, transit and release in a time course.

Q15 / C Cambial direction and living versus dead conducting cells

Answer: (C) P is inward secondary xylem; Q is outward secondary phloem.

Why: Vessel elements identify xylem and lose their protoplasts at functional maturity. Cambium makes secondary xylem inward and secondary phloem outward. Sieve-tube elements remain living despite losing a mature nucleus, functioning with companion cells, so absence of a nucleus alone does not equate them with vessels.

Why not the others: (A) reverses identity and direction; (B) wrongly makes mature vessels living; (D) assigns lateral-meristem derivatives to apical growth.

Key point: Combine production direction with the conducting cell's mature state.

Chemistry

Q16 / B Multi-functional connectivity, bond counts and carbon hybridisation

Answer: (B) (12,4;1,3,1).

Given: The neutral structure has five carbons, five hydrogens, two oxygens and one nitrogen; connectivity is shown explicitly.

Formula/Principle: Every bonded atom pair contributes one sigma bond; each double bond adds one pi and each triple bond two pi bonds.

Step 1: Seven heavy-atom connections contribute seven sigma bonds. Five C-H bonds bring the total to twelve.

Step 2:

$$N_\pi = 1_{\mathrm{C{=}C}} + 1_{\mathrm{C{=}O}} + 2_{\mathrm{C{\equiv}N}} = 4.$$

Step 3: The nitrile carbon is sp; the two alkene carbons and carbonyl carbon are sp2; the methylene between O and CN is sp3.

Why: Hybridisation follows the local sigma-domain geometry rather than the total count of bond lines. A triple bond has three bond components but still only one sigma connection. The oxygen-separated methylene is not part of the carbonyl's planar carbon centre.

Why not the others: (A) omits hydrogen sigma bonds; (C) loses a pi bond and assigns an extra sp carbon; (D) misclassifies the carbonyl carbon as sp3.

Key point: Count connections first, extra pi components second, and carbon geometries separately.

Q17 / A Mixture inversion from external chloride and ion counts

Answer: (A) 0.60 and 2.60.

Given: The first salt has two external chlorides; the second has one.

Formula/Principle: Immediate precipitation counts external chloride, while dissolved-ion count includes one complex ion per formula unit.

Step 1: Let the first salt's fraction be x.

$$2x + (1 - x) = 1.60 \Rightarrow x = 0.60.$$

Step 2: The first salt gives three ions and the second two.

$$n_{\mathrm{ions}} = 3(0.60) + 2(0.40) = 2.60 \text{ mol}.$$

Why: Coordinated chloride contributes to chemical composition but not to this immediate test.

Every formula unit also contributes a complex cation that is not counted by AgCl precipitation. The two measurements therefore use related but different counting rules.

Why not the others: (B) interchanges mixture fractions; (C) reports chloride ions alone as all ions; (D) mistakes 1.60/2 for the mixture fraction, ignoring the second salt's chloride.

Key point: Separate external-ion assays from total solution-ion counts.

Q18 / D Octet validity versus charge preference in CO Lewis proposals

Answer: (D) P gives both octets and formal charges C=-1,O=+1; Q leaves carbon short.

Formula/Principle: Formal charge equals valence electrons minus nonbonding electrons minus half the bonding electrons.

Step 1: In P, each atom counts six shared and two nonbonding electrons: both have octets.

Step 2:

$$FC_C = 4 - 2 - 3 = -1, \qquad FC_O = 6 - 2 - 3 = +1.$$

Step 3: Q gives carbon four shared plus two lone-pair electrons, only six; oxygen has eight.

Why: Correct total electron count is necessary but does not ensure each atom has an octet. The counterintuitive charge signs in P do not invalidate its electron bookkeeping. Formal charge and octet occupancy must be checked separately.

Why not the others: (A) overlooks carbon's six-electron shell in Q; (B) equates positive charge with an incomplete octet; (C) reverses the electron counts.

Key point: Count electrons around each atom before comparing formal charges.

Q19 / C Series limit versus a specified finite hydrogen transition

Answer: (C) 3.

Given: Balmer limit has final level 2 and initial level tending to infinity; Lyman-alpha is 2 to 1.

Formula/Principle: Ratios of wavelengths invert ratios of wavenumbers.

Step 1:

$$\lambda_B^{-1} = R/4 \Rightarrow \lambda_B = 4/R.$$

Step 2:

$$\lambda_L^{-1} = R(1 - 1/4) = 3R/4 \Rightarrow \lambda_L = 4/(3R).$$

Step 3:

$$\lambda_B/\lambda_L = 3.$$

Why: The limit is not the first Balmer line, whose initial level would be 3. Taking the inverse only after evaluating each energy difference avoids reversing the wavelength ratio. R cancels because both lines belong to hydrogen.

Why not the others: (A) reverses the ratio; (B) compares the Balmer limit with the Lyman limit using an incorrect factor; (D) omits the initial-level term in Lyman-alpha.

Key point: Identify both transition endpoints before using a series ratio.

Q20 / D Nitrogen-analysis stoichiometry and molecular atom count

Answer: (D) Two nitrogen atoms.

Given: Gas volume is dry, collection is quantitative and the molar volume already includes the measurement conditions.

Formula/Principle: Convert nitrogen gas to moles of nitrogen atoms, then divide by moles of compound.

Step 1:

$$n_{\mathrm{N_2}} = 0.224/22.4 = 0.0100 \text{ mol}, \quad n_N = 0.0200 \text{ mol}.$$

Step 2:

$$n_{\text{compound}} = 2.80/280 = 0.0100 \text{ mol}.$$

Step 3:

$$N \text{ atoms per molecule} = 0.0200/0.0100 = 2.$$

Why: N2 has two atoms per gas molecule, whereas the molecular count of the organic compound follows from its independently supplied molar mass. Comparing gas molecules directly with organic molecules loses that factor of two. No water-vapour correction is needed because the gas is explicitly dry.

Why not the others: (A) counts N2 molecules as atoms; (B) doubles the diatomic factor twice; (C) confuses the nitrogen mass percentage, 10%, with atom count.

Key point: Keep moles of molecules distinct from moles of elemental atoms.

Q21 / C Two-carbonyl reduction, product tests and stereocentre count

Answer: (C) Tollens negative; iodoform positive; two stereoisomers.

Step 1: The aldehyde becomes a primary benzyl alcohol, $-CH_2OH$, and the methyl ketone becomes $-CH(OH)CH_3$.

Step 2: No aldehyde remains for the ordinary Tollens test. The secondary alcohol still has the methyl-carbinol unit oxidisable to a methyl ketone, so the iodoform test remains positive.

Step 3: The ketone-derived carbon has four different substituents and gives an enantiomeric pair. The aldehyde-derived CH2OH carbon is not stereogenic.

Why: Reduction changes the functional tests differently: removing a carbonyl does not necessarily remove iodoform activity. Counting new stereocentres also requires distinguishing CHOH from CH2OH, rather than assigning a centre to every reduced carbonyl.

Why not the others: (A) leaves an aldehyde after stipulated complete reduction; (B) overlooks the methyl-carbinol response and stereocentre; (D) incorrectly treats CH2OH as a second stereocentre.

Key point: Reassess functional tests and stereochemistry on the product, not the starting material.

Q22 / B Hinsberg reaction versus product salt formation

Answer: (B) P primary, Q secondary, R tertiary.

Why: A primary amine leaves an N-H in the sulfonamide product, allowing formation of an alkali-soluble salt. A secondary amine's sulfonamide lacks that acidic N-H and stays insoluble. A tertiary amine cannot form that sulfonamide in the standard test but can protonate and dissolve in dilute acid.

Why not the others: (A) interchanges products with and without N-H; (C) assigns a sulfonamide to the tertiary amine; (D) makes the tertiary amine form Q's product and the secondary amine fail to react.

Key point: Interpret solubility through the product's remaining N-H, not only the starting amine.

Q23 / A Nucleic-acid identification from combined hydrolysis products

Answer: (A) Consistent with RNA.

Why: RNA contains ribose and normally uses uracil, whereas ordinary DNA contains deoxyribose and thymine. Phosphate alone does not discriminate because both nucleic-acid types contain it. The linked sugar and base evidence identifies a nucleic acid rather than a peptide or disaccharide.

Why not the others: (B) uses a shared component while ignoring the discriminating ones; (C) mistakes nitrogenous bases for amino acids; (D) cannot account for phosphate-linked nucleotides and bases.

Key point: Use the discriminating hydrolysis products rather than a component shared by candidates.

Q24 / D Piecewise-temperature first-order exposure

Answer: (D) 10 min.

Given: Let the T1 constant be k; the T2 constant is 2k and concentration is continuous at the switch.

Formula/Principle: Add first-order logarithmic conversion over consecutive intervals.

Step 1:

$$\ln(C_0/C_f) = 10k + 5(2k) = 20k.$$

Step 2:

$$20k = \ln 4 = 2\ln 2 \Rightarrow k = \ln 2/10 \text{ min}^{-1}.$$

Step 3:

$$t_{1/2,T1} = \ln 2/k = 10 \text{ min}.$$

Why: Changing temperature changes the subsequent rate, not the starting concentration anew. The two periods are equivalent to twenty minutes at T1, during which two half-lives occur. Summing logarithmic losses respects the multiplicative survival factors.

Why not the others: (A) gives the T2 half-life; (B) reports the equivalent total exposure as one half-life; (C) doubles it again rather than dividing by two halvings.

Key point: Add integrated rate exposures when a first-order constant changes between stages.

Q25 / A Selective precipitation onset and residual competing anion

Answer: (A) AgI first; iodide 10^-7 M.

Given: Initially both halides are dissolved; the two solubility products refer to free ions.

Formula/Principle: Precipitation begins when the relevant ion product reaches Ksp.

Step 1:

$$[\text{Ag}^+]_I = 10^{-16}/10^{-2} = 10^{-14} \text{ M}, \quad [\text{Ag}^+]_{Cl} = 10^{-10}/10^{-1} = 10^{-9} \text{ M}.$$

Step 2: AgI appears at the lower silver concentration. Until the AgCl threshold, chloride has not appreciably precipitated.

Step 3:

$$[\text{I}^-] = 10^{-16}/10^{-9} = 10^{-7} \text{ M}.$$

Why: Iodide is depleted while silver increases toward the chloride threshold. The final iodide concentration cannot be obtained by comparing initial concentrations alone; it follows from AgI equilibrium at the later common silver level.

Why not the others: (B) mishandles the exponent division; (C) gets the later iodide value but reverses the initial threshold; (D) ignores both threshold ordering and iodide removal.

Key point: Use separate onset thresholds, then a common-ion equilibrium at the second onset.

Q26 / B Comproportionation electron balance and diatomic iodine count

Answer: (B) 0.100 mol iodide and 0.060 mol iodine.

Given: I in iodate is +5, I in iodide is -1, and elemental iodine is zero.

Formula/Principle: Balance electron loss/gain, then atoms and charge.

Step 1: Reduction of one +5 iodine requires five electrons; five iodides each supply one electron.

Step 2:

$$\text{IO}_3^- + 5\,\text{I}^- + 6\,\text{H}^+ \longrightarrow 3\,\text{I}_2 + 3\,\text{H}_2\text{O}.$$

Step 3:

$$n_{\text{I}^-} = 5(0.020) = 0.100, \quad n_{\text{I}_2} = 3(0.020) = 0.060 \text{ mol}.$$

Why: All six iodine atoms, one from iodate and five from iodide, enter three diatomic molecules. Balancing only oxidation numbers without molecular iodine's factor of two gives the wrong product amount. The acid supplies hydrogen to form water and balances charge.

Why not the others: (A) ignores the five-electron requirement; (C) counts acid's coefficient as iodide; (D) counts iodine atoms as iodine molecules.

Key point: Follow electron balance through to molecular product stoichiometry.

Q27 / C Bond-inventory Hess estimate for hydrogenation

Answer: (C) -126 kJ/mol.

Given: Reactants have one C=C, four C-H and one H-H; product has one C-C and six C-H.

Formula/Principle: Estimated reaction enthalpy is total bond-breaking energy minus total bond-formation energy.

Step 1: Cancel the four C-H bonds common to both bond inventories.

Step 2:

$$\Delta H \approx (612 + 436) - (348 + 2 \times 413).$$

Step 3:

$$\Delta H \approx 1048 - 1174 = -126 \text{ kJ mol}^{-1}.$$

Why: Hydrogenation replaces the entire double-bond energy with a single bond and two extra C-H bonds. Counting all six product C-H bonds without also counting the initial four would destroy the cancellation. Average bond enthalpies give an estimate for gas-phase species, not an exact measured value.

Why not the others: (A) reverses breaking and formation; (B) omits formation of the product C-C bond; (D) counts an extra C-H bond as newly formed.

Key point: Compare complete bond inventories before cancelling unchanged bonds.

Q28 / D Liquid versus vapour composition from partial pressures

Answer: (D) 1/2 in liquid; 3/4 in vapour.

Given: Partial pressures sum to 40 kPa and the liquid obeys Raoult's law.

Formula/Principle: Liquid fraction uses pA/p*A; vapour fraction uses pA divided by total pressure.

Step 1:

$$x_A = 30/60 = 1/2.$$

Step 2:

$$y_A = 30/(30 + 10) = 3/4.$$

Why: The more volatile component is enriched in the vapour even though the liquid is equimolar. The two fractions use different denominators because one describes the liquid mixture and the other its gas phase. Equality of partial pressures to x times pure pressure does not make x equal to y.

Why not the others: (A) assumes equal phase compositions; (B) exchanges the fractions; (C) uses total gas pressure for both.

Key point: Identify the phase before choosing a mole-fraction denominator.

Q29 / A Ring opening inferred from one dialdehyde ozonolysis product

Answer: (A) Cyclohexene.

Step 1: The product has six carbons: two aldehyde carbons plus four methylenes.

Step 2: Each former alkene carbon becomes an aldehyde terminus. Reconnect those two termini by C=C while removing their carbonyl oxygens.

Step 3: The reconnecting bond closes a six-membered ring, giving cyclohexene with one double bond.

Why: Cutting a double bond in a ring can open it into one molecule with two carbonyl ends. An ordinary acyclic single alkene instead splits its skeleton into two disconnected products. The explicitly preserved internal chain length is essential, rather than the mere presence of aldehydes.

Why not the others: (B) yields separate methanal and pentanal; (C) contains two double bonds and gives additional fragments; (D) gives a five-carbon dialdehyde rather than six.

Key point: Rejoin carbonyl carbons and preserve the intervening chain when reconstructing cyclic alkenes.

Q30 / B Beta-hydrogen availability versus neighbouring SN2 hindrance

Answer: (B) No beta hydrogen and substantial neighbouring steric hindrance to SN2.

Why: The alpha carbon is CH2Br; its directly adjacent beta carbon is the central carbon bonded to three methyl groups and has no hydrogen. Methyl hydrogens are one bond farther away and cannot supply the required direct E2 beta hydrogen. The bulky neighbouring groups also impede backside approach despite the alpha carbon being primary.

Why not the others: (A) counts gamma hydrogens as beta; (C) proposes a rearranged carbon skeleton for a stipulated concerted pathway; (D) treats the primary label as sufficient while ignoring neighbouring steric crowding.

Key point: Locate alpha and beta atoms explicitly before applying reaction-class shortcuts.

Mathematics

Q31 / A Enumerating normals through an axial point and summing contact coordinates

Answer: (A) 8.

Given: Parametrise the parabola by $P(t) = (t^2, 2t)$.

Formula/Principle: A normal is perpendicular to the tangent at the point of contact; include the vertex separately when the tangent is vertical.

Step 1: For nonzero t the tangent slope is $1/t$, so the normal slope is $-t$. Its equation is $y = -tx+2t+t^3$. At t=0 the normal is y=0, also represented by this equation.

Step 2:

$$0 = -6t + 2t + t^3 = t(t^2 - 4) \Rightarrow t = 0, 2, -2.$$

Step 3:

$$\sum x(P_i) = 0^2 + 2^2 + (-2)^2 = 8.$$

Worked derivation:

$$P_i = (0,0), (4,4), (4,-4), \qquad \sum x_i = 8.$$

Why: Distinct normal contact points can have the same abscissa, so both off-axis contacts contribute. The vertex normal must not be discarded through division by t. Solving the normal-through-point condition enumerates all contacts rather than guessing one from symmetry.

Why not the others: (B) sums ordinates rather than abscissae; (C) counts only one off-axis contact abscissa; (D) adds the external point's abscissa to the contact sum.

Key point: Keep special tangent orientations and count contact points, not distinct x values.

Q32 / D Hyperbola scale from asymptotes and an off-axis point

Answer: (D) 10.

Given: Use $x^2/a^2 - y^2/b^2 = 1$, with positive a,b.

Formula/Principle: Asymptote slopes are $\pm b/a$; focal parameter satisfies $c^2 = a^2 + b^2$.

Step 1:

$$b = 2a.$$

Step 2: Substitute the given point:

$$9/a^2 - 16/(4a^2) = 1 \Rightarrow a^2 = 5, \quad b^2 = 20.$$

Step 3:

$$c = \sqrt{5 + 20} = 5, \qquad F_1F_2 = 2c = 10.$$

Worked derivation:

$$b^2 = 4a^2, \quad 9 - 4 = a^2, \quad 2\sqrt{a^2 + b^2} = 10.$$

Why: The asymptotes determine only the ratio of semiaxes, not the size. The point fixes that scale, after which the hyperbola's plus-sign focal relation applies. The requested separation is twice the centre-to-focus distance.

Why not the others: (A) reports c rather than 2c; (B) reports transverse-axis length 2a; (C) reports conjugate-axis length 2b.

Key point: Separate shape ratio, scale and focal separation.

Q33 / C Cubic root multiplicity versus corners of an absolute value

Answer: (C) a at or outside the two boundary values.

Given: Write $g(x) = x^3 - 3x + a$; possible corners of |g| occur at zeros of g.

Formula/Principle: A simple zero of g gives opposite nonzero one-sided slopes of |g|. A zero with g'=0 instead has derivative zero for |g|.

Step 1:

$$g'(x) = 3(x^2 - 1), \quad g(-1) = a + 2, \quad g(1) = a - 2.$$

Step 2: For -2<a<2, the local maximum is positive and local minimum negative. There are three simple zeros and hence three corners.

Step 3: For |a|>2 there is one simple real zero and hence one corner. At a=2 or -2 there is one double zero and one simple zero; the double zero is smooth after taking absolute value.

Worked derivation:

$$N_{\text{corners}} = \begin{cases} 3, & |a| < 2, \\ 1, & |a| \geq 2. \end{cases}$$

Why: Counting all polynomial roots as corners would mishandle the endpoint cases. Near a double zero, |g(x)| changes at order $(x - r)^2$, so its derivative at r is zero even if an absolute-value symbol remains in the expression. This distinction keeps both parameter endpoints.

Why not the others: (A) incorrectly excludes double-root boundary cases; (B) includes the three-corner interior; (D) selects precisely that interior.

Key point: Root multiplicity matters when an absolute value creates or removes a corner.

Q34 / B Complementary logarithmic integrands under reflection

Answer: (B) 1/2.

Given: On [0,1], both logarithm arguments are positive and their sum in the denominator is positive.

Formula/Principle: Reflect a definite integral by x=1-t and add it to the original.

Step 1:

$$I = \int_0^1 \frac{\ln(2 - x)}{\ln(2 - x) + \ln(1 + x)}\, dx.$$

Step 2:

$$2I = \int_0^1 \frac{\ln(1 + x) + \ln(2 - x)}{\ln(1 + x) + \ln(2 - x)}\, dx = \int_0^1 1\, dx = 1.$$

Step 3: Thus I=1/2.

Worked derivation:

$$f(x) + f(1 - x) = 1 \Rightarrow 2I = 1.$$

Why: Reflection exchanges the two numerator candidates while preserving the denominator. No elementary antiderivative for the original quotient is needed. Positivity on the closed interval ensures this addition never hides a zero denominator.

Why not the others: (A) reports the doubled integral; (C) integrates one logarithmic component as though the denominator were absent; (D) incorrectly treats the reflected pair as cancelling.

Key point: Test reflection symmetry before attempting an antiderivative.

Q35 / C Equivalence class as the full preimage of a cosine value

Answer: (C) Both signs of pi/3 modulo 2pi.

Given: Membership means cos x=cos(pi/3)=1/2.

Formula/Principle: Equality of cosine values gives both even-symmetry branches and periodic repetition.

Step 1:

$$\cos x = 1/2 \Rightarrow x = 2k\pi + \pi/3 \text{ or } x = 2k\pi - \pi/3.$$

Worked derivation:

$$[\pi/3]_R = \{2k\pi \pm \pi/3 : k \in \mathbb{Z}\}.$$

Why: Equality of images defines an equivalence relation, and its class contains every preimage of the same value. Cosine is not injective even within a full period, so retaining only the positive-angle branch loses valid members. Its period is 2pi, not pi for a fixed nonzero cosine value.

Why not the others: (A) omits the negative branch; (B) includes angles with cosine -1/2; (D) uses the wrong target cosine.

Key point: An equivalence class of a function value is its entire preimage.

Q36 / B Noncommuting products followed by trace/determinant reconstruction

Answer: (B) 2 and 3.

Given: Write $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

Formula/Principle: Matrix order matters; compute the two products before using trace and determinant.

Step 1:

$$AX - XA = \begin{pmatrix} 0 & -q \\ r & 0 \end{pmatrix} \Rightarrow q = -2,\ r = -3.$$

Step 2:

$$p + s = 5, \quad ps - qr = 0 \Rightarrow ps = 6.$$

Step 3:

$$p(5-p) = 6 \Rightarrow p^2 - 5p + 6 = 0 \Rightarrow p = 2, 3.$$

Worked derivation:

$$X = \begin{pmatrix} p & -2 \\ -3 & 5-p \end{pmatrix}, \quad (p-2)(p-3) = 0.$$

Why: The commutator constrains the off-diagonal entries but leaves the diagonal unconstrained. The remaining trace and determinant conditions then determine the two possible diagonal orders. Both values produce actual matrices satisfying all three requirements.

Why not the others: (A) corresponds to diagonal product -6, from a sign error; (C) wrongly sets diagonal product to zero; (D) gives product four rather than six.

Key point: Use each matrix constraint for the entries it actually controls.

Q37 / A Inferring trial bias then conditioning success parity

Answer: (A) 3/7.

Given: Nonzero p and 1-p allow cancellation in the equality.

Formula/Principle: Use binomial trial probabilities, then condition on exclusion of zero successes.

Step 1:

$$3p(1-p)^2 = 3p^2(1-p) \Rightarrow p = 1/2.$$

Step 2: Under the conditioning, an even success count can only be two.

$$P(N=2) = 3/8, \quad P(N \geq 1) = 1 - 1/8 = 7/8.$$

Step 3:

$$P(N \text{ even} \mid N \geq 1) = (3/8)/(7/8) = 3/7.$$

Worked derivation:

$$p = 1/2, \quad P_{\text{conditional}} = 3/7.$$

Why: The equality first determines the unknown trial bias. Zero is even but is removed by the stated conditioning, so it cannot enter the numerator. The event denominator changes along with this restriction.

Why not the others: (B) omits conditioning; (C) assumes even and odd are equally likely after excluding zero; (D) counts the remaining odd outcomes instead.

Key point: Infer a trial parameter before conditioning on a compound event.

Q38 / B Parameter inference followed by a signed coefficient sum

Answer: (B) 1.

Given: The first polynomial determines a; the second asks coefficient sum, not its value at x=0.

Formula/Principle: Binomial coefficients give individual terms; evaluating at x=1 sums coefficients.

Step 1:

$$\binom{5}{3} a^3 = 80 \Rightarrow 10a^3 = 80 \Rightarrow a = 2.$$

Step 2:

$$\sum c_k = (1-a)^4 = (1-2)^4 = 1.$$

Worked derivation:

$$a = 2, \quad Q(1) = (-1)^4 = 1.$$

Why: A real cube root is unique, so the parameter sign is fixed by the positive coefficient. The minus sign in the second polynomial makes the coefficient sum different from $(1+a)^4$. Individual coefficients can be negative even though their sum here is positive.

Why not the others: (A) substitutes a plus sign; (C) omits the constant one before raising to the fourth power; (D) forgets the even exponent.

Key point: Distinguish extracting a coefficient from evaluating a polynomial to sum coefficients.

Q39 / C AP block averages and a least-index term threshold

Answer: (C) 14.

Given: Let the first term be a and common difference d>0.

Formula/Principle: The sum of an odd-length AP block is its length times its middle term.

Step 1:

$$5(a+2d)=40, \quad 5(a+7d)=115.$$

Step 2:

$$5d=23-8=15 \Rightarrow d=3, \quad a=8-6=2.$$

Step 3:

$$a_n=2+3(n-1)=3n-1>40 \Rightarrow n>41/3.$$

The least integer is 14.

Worked derivation:

$$a_{13}=38, \quad a_{14}=41.$$

Why: The given blocks constrain term averages, not cumulative sums up to ten. Once the AP is reconstructed, the strict term threshold requires an integer endpoint check. A term equal to or below 40 does not meet the condition.

Why not the others: (A) gives 35; (B) gives 38; (D) gives 44 but is not least.

Key point: Reconstruct a progression before applying a strict index threshold.

Q40 / D Vanishing amplitude despite a sign oscillation

Answer: (D) 0.

Given: x is positive near the limit and the sign factor always belongs to {-1,0,1}.

Formula/Principle: Bound the entire product rather than seeking a limit of each factor separately.

Step 1:

$$-x \le x\,\mathrm{sgn}(\sin(1/x)) \le x.$$

Step 2: Both bounding functions tend to zero from the right.

Worked derivation:

$$|x\,\mathrm{sgn}(\sin(1/x))| \le x \to 0.$$

Why: The sign factor oscillates, but its magnitude never exceeds one. Multiplication by a quantity shrinking to zero bounds the full oscillation amplitude. A product can therefore have a limit even when one factor has no limit.

Why not the others: (A) treats the sign as constantly positive and omits x's decay; (B) makes the analogous negative mistake; (C) considers the oscillation without its vanishing amplitude.

Key point: An oscillating bounded factor may be controlled by a vanishing multiplier.

Q41 / A Common-perpendicular endpoints and midpoint

Answer: (A) (1/2,1/2,1).

Given: Let P=(t,t,0), Q=(1+s,-s,2). Direction vectors are (1,1,0) and (1,-1,0).

Formula/Principle: The shortest joining vector of nonparallel skew lines is perpendicular to both directions.

Step 1:

$$(Q-P)\cdot(1,1,0)=1-2t=0 \Rightarrow t=1/2.$$

Step 2:

$$(Q-P)\cdot(1,-1,0)=1+2s=0 \Rightarrow s=-1/2.$$

Step 3:

$$P = (1/2, 1/2, 0), \quad Q = (1/2, 1/2, 2), \quad (P+Q)/2 = (1/2, 1/2, 1).$$

Worked derivation:

$$\overrightarrow{PQ} = (0,0,2) \perp (1,1,0), (1,-1,0).$$

Why: A connecting segment must meet both perpendicular conditions, not merely join arbitrarily chosen parameter-zero points. The two direction vectors are independent, so these conditions determine the unique pair of nearest points. Taking their average then answers the requested midpoint rather than only the distance.

Why not the others: (B) reverses the y sign of both feet; (C) uses a parameter-zero coordinate; (D) averages the line origins without checking perpendicularity.

Key point: Find both feet before reporting a common-perpendicular location.

Q42 / D Circular separation plus a second excluded-neighbour condition

Answer: (D) 24.

Given: Fix A at seat 0 and number the other seats clockwise 1 through 5. B must be at 2 or 4.

Formula/Principle: Fix one person to remove rotations, then count remaining allowed positions without quotienting reflections.

Step 1: There are two positions for B.

Step 2: C may occupy seats 2,3,4 but not B's occupied seat, leaving two choices for each B placement.

Step 3: The other three people fill the remaining seats in 3! ways.

$$N = 2 \times 2 \times 3! = 24.$$

Worked derivation:

$$N_{\text{rotations identified}} = 24.$$

Why: The separation constraint is not an adjacency block: two possible clockwise placements of B are valid. The C restriction must then exclude both neighbours of A and B's already occupied seat. Reflected arrangements remain distinct, so no further factor of two is removed.

Why not the others: (A) incorrectly identifies reflections; (B) leaves C three options despite B occupying one; (C) ignores the excluded seats next to A.

Key point: Resolve interacting seat constraints after fixing the rotational reference.

Q43 / C Principal-argument wrapping under squaring

Answer: (C) (pi/2,pi].

Given: Write z=exp(i theta) with -pi<theta<=pi.

Formula/Principle: Reduce the doubled angle to the stated principal interval.

Step 1: For -pi<2theta<=pi, no 2pi subtraction is permitted or needed.

Step 2: Subtraction by 2pi is required exactly when

$$2\theta > \pi \Rightarrow \theta > \pi/2.$$

Step 3: The original upper bound theta<=pi remains, and theta=pi gives Arg(z squared)=0, correctly.

Worked derivation:

$$\text{Arg}(z^2) = 2\theta - 2\pi \quad \Longleftrightarrow \quad \pi/2 < \theta \leq \pi.$$

Why: At theta=pi/2 the squared argument is pi, which is already allowed and must not be replaced by -pi. For theta below -pi/2 the correction would be addition, not subtraction, of 2pi. The endpoints depend on the chosen principal interval.

Why not the others: (A) selects the no-wrap region and wrong endpoint; (B) requires the opposite wrap sign; (D) is the central no-wrap region.

Key point: Apply branch correction only after locating the unreduced angle.

Q44 / D Separable initial value with branch and finite-pole interval

Answer: (D) 4/3 and (-ln2,infinity).

Given: The initial value is not either equilibrium solution 0 or 1.

Formula/Principle: Separate variables, integrate partial fractions and exclude any pole of the resulting solution.

Step 1:

$$\int \frac{dy}{y(1-y)} = x + C \Rightarrow \ln\left|\frac{y}{1-y}\right| = x + C.$$

Step 2: The initial value selects

$$y = \frac{2e^x}{2e^x - 1}.$$

Step 3:

$$y(\ln 2) = 4/3, \qquad 2e^x - 1 = 0 \Longleftrightarrow x = -\ln 2.$$

Step 4: The largest pole-free interval containing zero is (-ln2,infinity).

Worked derivation:

$$y' = -\frac{2e^x}{(2e^x - 1)^2} = y(1-y), \quad y(0) = 2.$$

Why: The separated integration determines a branch through the initial point, not a function that can be continued across an infinite pole. A correct value at ln2 alone does not establish a global domain. Substitution verifies both the differential equation and the initial condition.

Why not the others: (A) crosses the pole; (B) uses the wrong solution branch/value; (C) places the excluded pole at the wrong sign.

Key point: An initial-value solution includes its connected domain, not only its formula.

Q45 / A Doubled-angle inverse compositions with four branch regions

Answer: (A) x>=1.

Given: Both inverse-function arguments lie in [-1,1] for real x. Let $\theta = 2\arctan x \in (-\pi, \pi)$.

Formula/Principle: The arguments are sin theta and cos theta, but inverse functions return principal representatives.

Step 1:

$$\arccos(\cos\theta) = |\theta|.$$

Step 2: For nonnegative angles the sum is

$$\begin{cases} 2\theta, & 0 \le \theta \le \pi/2, \\ \pi, & \pi/2 \le \theta < \pi. \end{cases}$$

The first branch reaches π only at its right endpoint; the second equals π throughout.

Step 3: For negative angles the sum is

$$\begin{cases} 0, & -\pi/2 \le \theta < 0, \\ -\pi - 2\theta, & -\pi < \theta < -\pi/2. \end{cases}$$

Both branches are strictly less than π.

Step 4: Therefore $\theta \ge \pi/2$, equivalent to $x \ge \tan(\pi/4) = 1$.

Worked derivation:

$$\theta \in [\pi/2, \pi) \Longleftrightarrow x \in [1, \infty).$$

Why: Replacing arcsin(sin theta) by theta on its entire doubled-angle domain would discard a whole valid branch. Negative theta cannot reach equality at finite x; its apparent limiting equality would require theta=-pi, excluded by arctangent's range. x=1 is genuinely valid.

Why not the others: (B) drops the valid endpoint; (C) incorrectly mirrors the positive solution across zero; (D) keeps the wrong side of the branch boundary.

Key point: Reduce inverse-trigonometric compositions on their actual principal branches.

Physics

Q46 / D Backward inference of projectile components from height and direction

Answer: (D) $4\sqrt{10}$ m.

Given: Vertical motion has constant acceleration -g; horizontal velocity u_x is constant.

Formula/Principle: Maximum height fixes the initial vertical component; the measured direction links the instantaneous components.

Step 1:

$$u_y^2 = 2gH = 100 \Rightarrow u_y = 10 \text{ m s}^{-1}.$$

Step 2: At height 3 m,

$$v_y^2 = u_y^2 - 2g(3) = 40.$$

Upward motion selects positive v_y. Since tan45=1, $u_x = v_y = 2\sqrt{10}$ m/s.

Step 3:

$$T = 2u_y/g = 2 \text{ s}, \quad R = u_x T = 4\sqrt{10} \text{ m}.$$

Worked derivation:

$$R = \sqrt{2g(H-3)}\,2\sqrt{2H/g} = 4\sqrt{10} \text{ m}.$$

Why: The 45-degree direction is specified at an intermediate height, not at launch. It therefore relates u_x to the reduced vertical speed at that height. Using the initial vertical component for that direction would overestimate range.

Why not the others: (A) reports horizontal speed as a distance; (B) incorrectly makes launch components equal; (C) reports initial vertical speed numerically rather than range.

Key point: Apply a measured velocity direction at the time and height where it was measured.

Q47 / C Friction threshold with independent vertical unloading

Answer: (C) 12 N.

Given: Vertical acceleration is zero, and the upward force acts separately from the horizontal pull.

Formula/Principle: First find the actual normal reaction, then use limiting static friction.

Step 1:

$$N + 6 - mg = 0 \Rightarrow N = 30 - 6 = 24 \text{ N}.$$

Step 2:

$$F_{\text{threshold}} = \mu_s N = 0.50(24) = 12 \text{ N}.$$

Worked derivation:

$$F_{\text{threshold}} = \mu_s(mg - F_{\text{up}}) = 12 \text{ N}.$$

Why: Friction is limited by contact force, not weight in every arrangement. The vertical support unloads the floor even though the block remains in contact. Static friction adjusts below its limit until the applied horizontal force reaches this threshold.

Why not the others: (A) uses N=mg; (B) adds the upward force to weight before multiplying by friction coefficient; (D) reports weight, not limiting friction.

Key point: Resolve forces normal to a surface before using a friction coefficient.

Q48 / B Terminal force balance followed by fixed-power transient acceleration

Answer: (B) 12 m/s and 1.5 m/s squared.

Given: Power is constant and force remains parallel to positive velocity.

Formula/Principle: Engine force is P/v, not constant; terminal speed requires zero net force.

Step 1:

$$P/v_T = f \Rightarrow v_T = 36/3 = 12 \ \mathrm{m\,s^{-1}}.$$

Step 2: At v=6 m/s,

$$F_{\text{engine}} = 36/6 = 6 \ \mathrm{N}.$$

Step 3:

$$a = (6-3)/2 = 1.5 \ \mathrm{m\,s^{-2}}.$$

Worked derivation:

$$a(v_T/2) = \frac{2f - f}{m} = \frac{f}{m} = 1.5 \ \mathrm{m\,s^{-2}}.$$

Why: A fixed-power drive provides a larger force at lower speed. Resistance must be subtracted before dividing by mass; engine power alone is not net power. The model is used at positive nonzero speed, so no startup singularity is invoked.

Why not the others: (A) mistakes the half-terminal measurement speed for terminal speed; (C) omits resistance in acceleration; (D) makes both errors.

Key point: Distinguish constant power, driving force and net force.

Q49 / C Counterrotating inelastic coupling and energy loss

Answer: (C) $-\omega/2$; $9/10$ lost.

Given: Positive and negative signs refer to the same fixed axis; the final inertia is 2I.

Formula/Principle: Angular momentum is conserved in coupling, but kinetic energy need not be.

Step 1:

$$L_i = I\omega - 2I\omega = -I\omega = (2I)\Omega \Rightarrow \Omega = -\omega/2.$$

Step 2:

$$K_i = \tfrac{1}{2}I\omega^2 + \tfrac{1}{2}I(2\omega)^2 = \tfrac{5}{2}I\omega^2, \quad K_f = \tfrac{1}{2}(2I)(\omega/2)^2 = \tfrac{1}{4}I\omega^2.$$

Step 3:

$$1 - K_f/K_i = 1 - 1/10 = 9/10.$$

Worked derivation:

$$\Omega = -\omega/2, \quad K_f/K_i = 1/10.$$

Why: Opposite rotation reduces signed momentum without cancelling positive kinetic energies. The sticking constraint fixes the final common speed, while internal friction converts the missing mechanical energy to other forms. No external torque does not imply no kinetic-energy loss.

Why not the others: (A) fails to include both discs in final inertia; (B) loses the net momentum sign; (D) reports retained rather than lost fraction.

Key point: Add angular momenta with signs and energies as nonnegative quantities.

Q50 / B Mean planetary density from a surface-orbit period

Answer: (B) 3pi/(G T squared).

Given: The orbit radius is approximated by planetary radius R, and mean density is M divided by spherical volume.

Formula/Principle: Equate gravitational acceleration and circular centripetal acceleration, then eliminate M/R cubed.

Step 1:

$$GM/R^2 = (2\pi/T)^2 R \Rightarrow T^2 = 4\pi^2 R^3/(GM).$$

Step 2:

$$M = (4\pi/3)\rho R^3.$$

Step 3:

$$T^2 = 3\pi/(G\rho) \Rightarrow \rho = 3\pi/(GT^2).$$

Worked derivation:

$$\rho = \frac{3\pi}{GT^2}, \quad [GT^2]^{-1} = \mathrm{kg\,m^{-3}}.$$

Why: The radius cancels because both mass at fixed density and the orbital relation contain R cubed. The result is a mean density, not a local surface density. A substantial orbital altitude would invalidate replacing the orbit radius by R and require an extra geometric factor.

Why not the others: (A) omits the factor three from sphere volume; (C) introduces an extra factor two; (D) carries the orbital 4pi-squared factor through without dividing by spherical volume.

Key point: Eliminate an unmeasured scale using independent geometry and dynamics relations.

Q51 / C Siphon kinetic head and crest vapour-pressure limit

Answer: (C) 4 m.

Given: Outlet and reservoir surface are at atmospheric pressure; the tube has the same speed throughout.

Formula/Principle: Apply Bernoulli first from surface to outlet, then from surface to crest.

Step 1:

$$v^2/(2g) = 5.0\ \mathrm{m}.$$

Step 2: In water-head units at the crest,

$$p_c/(\rho g) = 10.0 - H - v^2/(2g) = 5.0 - H.$$

Step 3: Set the limiting pressure head to 1.0 m:

$$5.0 - H = 1.0 \Rightarrow H = 4.0\ \mathrm{m}.$$

Worked derivation:

$$H_{\text{limit}} = h_{\text{atm}} - h_{\text{vap}} - h_{\text{outlet drop}} = 4\ \mathrm{m}.$$

Why: The moving water requires kinetic head as well as elevation head. Ignoring that speed would overestimate the allowable crest height. Vapour pressure, not zero absolute pressure, is the stipulated lower-pressure limit; real losses or other constraints can reduce the practical height further.

Why not the others: (A) sets the pressure limit to zero; (B) ignores kinetic head; (D) ignores both vapour pressure and tube speed.

Key point: Determine flow speed before imposing a minimum-pressure condition at a high point.

Q52 / D Heat-capacity accounting over a complete P-V cycle

Answer: (D) 2/13.

Given: Heat input occurs on A-D and D-C; the other legs cool the gas.

Formula/Principle: For each leg use $\Delta U = (3/2)\Delta(PV)$ and $Q = \Delta U + W$; net work is the signed loop area.

Step 1: A-D is isochoric:

$$W_{AD} = 0, \quad Q_{AD} = (3/2)(200 - 100) = 150 \text{ J}.$$

Step 2: D-C is isobaric:

$$W_{DC} = 200 \text{ J}, \quad \Delta U_{DC} = (3/2)(400 - 200) = 300 \text{ J}, \quad Q_{DC} = 500 \text{ J}.$$

Step 3: Compression B-A does -100 J work; C-B does zero work.

$$W_{\text{net}} = 200 - 100 = 100 \text{ J}.$$

Step 4:

$$\eta = 100/(150 + 500) = 100/650 = 2/13.$$

Worked derivation:

$$Q_{\text{out}} = 550 \text{ J}, \quad Q_{\text{in}} - Q_{\text{out}} = 100 \text{ J} = W_{\text{net}}.$$

Why: Heat input is not confined to the expansion leg; heating at fixed volume also supplies energy. Conversely, positive expansion work is not the cycle's net work because compression returns some of it. A cyclic internal-energy sum is zero, providing an independent heat-work balance.

Why not the others: (A) omits isochoric heat input; (B) omits compression work; (C) omits expansion work from heat input while retaining internal-energy increases.

Key point: Sum heat and work by leg before computing a cyclic efficiency.

Q53 / A Charge-sign reversal of potential-graph stability

Answer: (A) x=0 only.

Given: For charge q<0, potential energy is U=qV; motion is restricted to x.

Formula/Principle: Equilibrium requires zero potential slope; stability requires a local minimum of U, not necessarily V.

Step 1:

$$\frac{dV}{dx} = \frac{4V_0 x}{a^2}\left(\frac{x^2}{a^2} - 1\right) = 0 \Rightarrow x = 0, \pm a.$$

Step 2: The graph has a local maximum at zero and minima at +/-a. Multiplication by negative q reverses those energy classifications.

Step 3: Thus U has its local minimum at zero only.

Worked derivation:

$$U''(0) = q(-4V_0/a^2) > 0, \qquad U''(\pm a) = q(8V_0/a^2) < 0.$$

Why: A negative charge is driven toward higher electrical potential because its energy is charge times potential. Zero field identifies all three equilibria but does not establish which are stable. The one-dimensional restriction avoids any unsupported claim about stability in other spatial directions.

Why not the others: (B) uses the positive-charge rule; (C) confuses equilibrium with stability; (D) ignores the energy minimum at zero.

Key point: Convert potential to charge-specific potential energy before judging stability.

Q54 / B Opposite-polarity cancellation of unequal capacitors' equal charges

Answer: (B) Zero.

Given: Each capacitor initially has charge magnitude CV; the cross-connection brings opposite signs onto each common node.

Formula/Principle: Conserve total charge on each isolated final node, including signs.

Step 1:

$$Q_1 = CV, \quad Q_2 = 2C(V/2) = CV.$$

Step 2: One node receives $+Q_1 - Q_2 = 0$; the other receives the opposite zero net charge.

Step 3:

$$V_f = (Q_1 - Q_2)/(C + 2C) = 0.$$

Worked derivation:

$$Q_{\text{node}} = 0 \Rightarrow V_f = 0.$$

Why: Capacitance and voltage differ but the stored charge magnitudes are equal. Opposite-polarity connection cancels node charges, unlike positive-to-positive connection. Initial field energy can dissipate in the specified resistive wires, so energy need not remain as a final capacitor voltage.

Why not the others: (A) assumes simple voltage averaging; (C) uses only one initial charge; (D) adds equal charge magnitudes without their signs.

Key point: Reconnection problems require signed node charge, not unsigned stored charge.

Q55 / A Central-field cancellation versus nonzero magnetic moment

Answer: (A) J=2I and total moment -7I pi R squared along z.

Given: Central loop fields scale as current/radius, while moments scale as current times area.

Formula/Principle: Superpose signed centre fields first; calculate moments separately using their different geometric dependence.

Step 1:

$$\mu_0 I/(2R) - \mu_0 J/(4R) = 0 \Rightarrow J = 2I.$$

Step 2:

$$m_z = I\pi R^2 - J\pi(2R)^2 = (1-8)I\pi R^2 = -7I\pi R^2.$$

Worked derivation:

$$B_z(0) = 0 \quad \text{but} \quad m_z = -7I\pi R^2.$$

Why: Cancellation at one observation point does not require cancellation of the system's magnetic moment. Larger radius reduces centre field per unit current but increases area per unit current. The opposing loop therefore needs more current and dominates the total moment.

Why not the others: (B) equates field cancellation with moment cancellation; (C) reverses radius scaling in the centre field; (D) uses area scaling for the field condition.

Key point: Different magnetic quantities can weight the same current geometry differently.

Q56 / D Off-centre rotation with two arm potentials and polarity

Answer: (D) $3B\omega a^2/2$; A higher.

Given: End distances from the pivot are a and 2a; the opposite arms rotate in the same rigid motion.

Formula/Principle: Motional charge separation gives potential gradient $dV/dr = (v \times B)_r = -B\omega r$ for the stated directions.

Step 1: For each arm,

$$V(r) - V(0) = -\int_0^r B\omega s\, ds = -B\omega r^2/2.$$

Step 2:

$$V_B - V_A = -B\omega[(2a)^2 - a^2]/2 = -3B\omega a^2/2.$$

Step 3: The negative sign makes B lower than A; the emf magnitude is its absolute value.

Worked derivation:

$$|\mathcal{E}_{AB}| = 3B\omega a^2/2, \quad V_A > V_B.$$

Why: Positive magnetic force points toward the pivot on both arms, so both outer ends are below the pivot. Their potentials must be subtracted rather than adding two outward emf magnitudes. Treating the full 3a rod as rotating about an end would use the wrong speed distribution.

Why not the others: (A) uses the full length as an end-pivot radius; (B) reverses polarity; (C) adds the two arm magnitudes instead of subtracting potentials.

Key point: For an off-centre rotating rod, compare each endpoint with the common pivot.

Q57 / A Prism emergence boundary and independent minimum-deviation path

Answer: (A) (sqrt6-1)/2 and 60 degrees.

Given: At the second face the critical angle c satisfies sin c=1/sqrt3. The requested incidence threshold is the grazing-emergence limit.

Formula/Principle: Combine the internal angle sum with Snell's law at entry; for minimum deviation use the symmetric internal path.

Step 1: At threshold,

$$r_2 = c, \quad r_1 = 60^\circ - c, \quad \sin i_{\text{min}} = \sqrt{3}\sin(60^\circ - c).$$

Step 2: With $\cos c = \sqrt{2/3}$,

$$\sin i_{\text{min}} = \sqrt{3}\left(\frac{\sqrt{3}}{2}\sqrt{\frac{2}{3}} - \frac{1}{2\sqrt{3}}\right) = \frac{\sqrt{6}-1}{2}.$$

Step 3: Minimum deviation has r1=r2=30 degrees, giving $\sin i = \sqrt{3}/2$, hence i=e=60 degrees.

Step 4:

$$\delta_{\text{min}} = i + e - A = 60 + 60 - 60 = 60^\circ.$$

Worked derivation:

$$\sin i_{\text{threshold}} = (\sqrt{6}-1)/2, \quad \delta_{\text{min}} = 60^\circ.$$

Why: Incidence at the first face and incidence at the second are not equal except on the symmetric path. A smaller entry angle gives a larger second internal angle and eventually TIR. The emergence threshold and minimum-deviation condition are separate constraints and must not be merged.

Why not the others: (B) uses the symmetric internal angle as the external threshold; (C) confuses an internal half-apex angle with deviation; (D) uses the wrong sign in the angle-difference identity and exceeds a physical sine.

Key point: Apply boundary refraction and symmetric deviation as distinct parts of prism analysis.

Q58 / B Optimisation of transmission through an inserted polariser

Answer: (B) I0/8 at 45 degrees.

Given: The first polariser halves unpolarised intensity; the final transmission axis is 90 degrees from the first.

Formula/Principle: Apply Malus's law consecutively to the polarised beams.

Step 1:

$$I_1 = I_0/2.$$

Step 2:

$$I_f = (I_0/2)\cos^2\theta\cos^2(90^\circ - \theta) = (I_0/8)\sin^2(2\theta).$$

Step 3: The maximum is I0/8 when sin squared(2theta)=1, for example theta=45 degrees.

Worked derivation:

$$I_{\text{max}} = I_0/8.$$

Why: The intermediate polariser changes the polarisation direction before the light reaches the last one. Crossed first and last axes therefore do not force zero transmission when another projection intervenes. Each projection also loses intensity, so rotating the beam does not transmit the full initially polarised intensity.

Why not the others: (A) ignores the intermediate change of direction; (C) misses one factor of one half at the optimum; (D) ignores both crossed-axis rejection and projection losses.

Key point: Apply polarisation projections in optical order.

Q59 / C Off-resonance photon energy between adjacent absorption lines

Answer: (C) No resonant bound excitation or ionisation occurs under these conditions.

Given: Initial energy is -3.40 eV and final bound energy after absorption would have to be -1.40 eV.

Formula/Principle: Bound absorption requires an exact allowed energy difference; ionisation requires at least the continuum threshold.

Step 1:

$$\Delta E_{2\to3} = 1.89 \text{ eV}, \quad \Delta E_{2\to4} = 2.55 \text{ eV}.$$

Step 2: 2.00 eV lies strictly between these adjacent gaps, and all higher gaps are larger. Ionisation needs 3.40 eV.

Worked derivation:

$$1.89 < 2.00 < 2.55 < 3.40 \quad (\text{eV}).$$

Why: An isolated bound transition does not absorb an arbitrary excess and discard the remainder while ending in n=3. The supplied exclusions remove broadening, collision-assisted and multiphoton alternatives. Positive photon energy alone does not imply sufficient energy for the continuum.

Why not the others: (A) treats a discrete transition like a threshold; (B) ignores binding energy; (D) rounds an exact resonance condition.

Key point: Distinguish discrete excitation from continuum ionisation.

Q60 / D Fusion binding change and two-body recoil partition

Answer: (D) 2.55 MeV.

Given: Initial bound nucleons total four; final helium-3 has three and the free neutron has zero binding energy.

Formula/Principle: Reaction energy is final minus initial total binding; opposite equal momenta divide kinetic energy inversely with final masses.

Step 1:

$$Q = 3(2.60) - 2[2(1.10)] = 7.80 - 4.40 = 3.40 \text{ MeV}.$$

Step 2:

$$K_n/K_{\text{He}} = m_{\text{He}}/m_n = 3.$$

Step 3:

$$K_n = (3/4)Q = 2.55 \text{ MeV}.$$

Worked derivation:

$$K_n + K_{\text{He}} = 3.40, \quad K_n = 3K_{\text{He}}.$$

Why: Per-nucleon binding must first be multiplied by each nucleus's nucleon count. Not all released energy belongs to the neutron because momentum conservation also makes helium recoil. The lighter product receives the larger share for equal momenta.

Why not the others: (A) gives all Q to the neutron; (B) gives helium's one-quarter share; (C) subtracts per-nucleon numbers without nuclear counts or recoil.

Key point: Calculate total binding change before distributing two-body recoil energy.